

LABORATORY SESSION 2

Cell and Tissue Injury

Cellular Swelling · Fatty Change · Hyalinization · Muroid Change · Calcification

PART 1 The Normal Baseline — Liver and Gut

PART 2 Reversible Injury — the Five Changes

PART 3 Gross Specimens

PART 4 Microscopic Slides

PART 5 Homework, Self-Test and Glossary

SYNTHESIZED FROM

“Pathological Experiment-2: Cell and Tissue Injury”

Pathology Institute, School of Medicine, Zhejiang University · 38 slides

**Robbins & Kumar *Basic Pathology*, 11e (Ch. 1–2) · Robbins & Cotran *Pathologic Basis of Disease*
· Wheater’s *Functional Histology***

For the mechanisms the slides state but do not explain

How to Use This Guide

Session 1 was about *adaptation* — a cell changing successfully. This session is about what happens when the change is no longer enough and the cell begins to fail, but has not yet died. Every lesion here is **reversible**: remove the cause and the cell recovers. That single word organises the whole session.

The organising idea

Four of the five changes are a **substance accumulating where it should not be**, and the fifth is water. Ask of every specimen: *what has built up, and why could the cell not get rid of it?* Water builds up when the sodium pump fails. Fat builds up when the liver cannot export it. Protein builds up as hyaline. Mucin builds up in connective tissue. Calcium builds up in tissue that is already dead. Five answers, one question.

The two tags

As in Session 1, every observation is marked for where it comes from:

- **GROSS** — naked-eye examination: size, weight, shape, colour, surface, cut surface.
- **MICRO** — down the microscope: architecture at 4×, cells at 40×.

BENCH TECHNIQUE — THE METHOD IS IN SESSION 1

The description templates — **organ → whole organ → lesion** for a gross specimen, **naked eye → 4× → 40×** for a slide — and the department's four report headings are all in the Session 1 guide, Part 1. They are not repeated here. If you have not internalised them yet, read that Part again before the bench; nothing in this session changes the method.

The five kinds of box

Box	What it contains
High-Yield	Repeatedly examined. If you memorise nothing else, memorise these.
Exam Trap	Where two lesions are confused, or the intuitive answer is wrong.
Bench Technique	How to actually do it: which objective, which stain, what to write down.
Clinical Correlation	What the patient with this specimen looked like in life.
Mechanism	A numbered chain in which each step forces the next, with the <i>why</i> spelled out.

Suggested study sequence

1. **Before the lab (≈30 min).** Part 2. Go in knowing the five changes and what accumulates in each.
2. **At the bench.** Part 1 open beside the microscope — you cannot call a hepatocyte swollen until you know how big a normal one is.
3. **Writing the report (≈45 min).** Parts 3 and 4, one card per specimen. Your homework is two of them.
4. **Night before the exam (≈20 min).** Part 5 — the look-alike tables, the self-test, then the glossary.

THE THREE THINGS THE LECTURER SAYS TO MASTER

Part two of the deck states its purpose in three lines, and they are worth treating as the examinable core of the whole session:

- Master the morphological characters of **reversible injury** — hyaline degeneration, mucoid change and pathological calcification.
- **Distinguish cellular swelling from fatty change.**
- **Identify the tigroid heart and the fat heart.**

The second and third are *distinguish* and *identify* tasks, not definitions — which is why §5.3 gives each of them its own comparison table.

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PART ONE

01

The Normal Baseline — Liver and Gut

Almost every lesion in this session is in the liver, and the rest are read against it. A swollen hepatocyte, a fat-filled hepatocyte and a normal one differ by details you can only see if you already know what the normal cell looks like — so this Part comes first, and it belongs beside the microscope rather than in your memory.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- State the normal weight, site, colour and functions of the liver.
- Identify central vein, liver plates, sinusoids and portal space on a slide.
- Name the four layers of the digestive tube and what each contains.

1.1 Liver — Gross

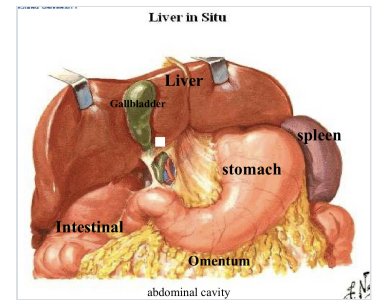


Fig 1.1 The liver in situ, with the gallbladder beneath it and the stomach, spleen, intestine and omentum around it. Its relations matter for description: a liver specimen usually arrives with the gallbladder bed and porta hepatis visible, and those are your landmarks for orientation.

TABLE 1.1 NORMAL LIVER — THE SIX WORDS

Feature	Normal
Weight	Male 1230–1446 g • Female 1100–1300 g — the heaviest internal organ
Site	Abdominal cavity — right upper quadrant, under the diaphragm
Colour	Red, and dull — not glistening. Both halves of that phrase matter: pallor and loss of lustre are the two colour changes you will meet in §3.1
Shape and edges	Wedge-shaped, with a sharp anterior edge . Remember this . A blunt, rounded edge is abnormal and appears in three of this session's specimens
Capsule	Thin, smooth, not tense — Glisson's capsule
Function	The biggest glandular organ — a “ chemical plant ”: metabolism, bile formation, detoxification, blood clotting, immunisation, calorogenesis, and regulation of water and electrolytes

HIGH-YIELD — WHY THE LIVER IS THE ORGAN OF THIS SESSION

Look at the function list again. The liver **metabolises** (so it is exposed to every toxin absorbed from the gut), it **detoxifies** (so it takes the hit), and it is the body's main site of **lipid handling** (so it is where fat accumulates when handling fails). A “chemical plant” working on that scale is the first organ to show a metabolic insult — which is why three of this session's seven gross specimens are liver.

1.2 Liver — Histology

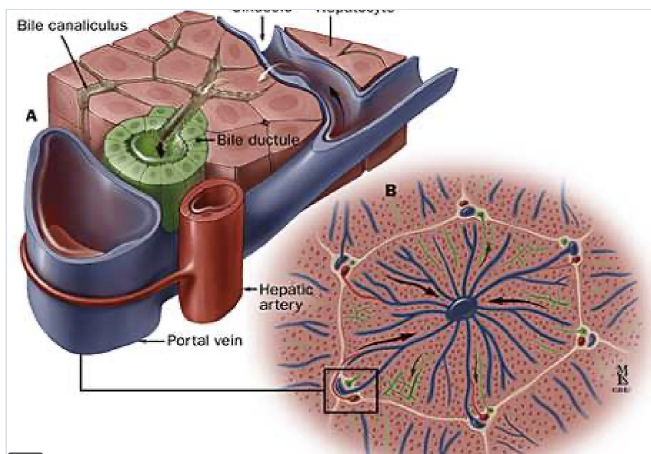


Fig 1.2 The lobule in three dimensions. **A:** hepatocytes in plates, with a **bile canaliculus** between adjacent cells and a **sinusoid** running alongside. **B:** the hexagonal lobule — blood flows inwards from the portal triads to the central vein; bile flows outwards to the bile ductules. Two fluids, opposite directions, same plate.

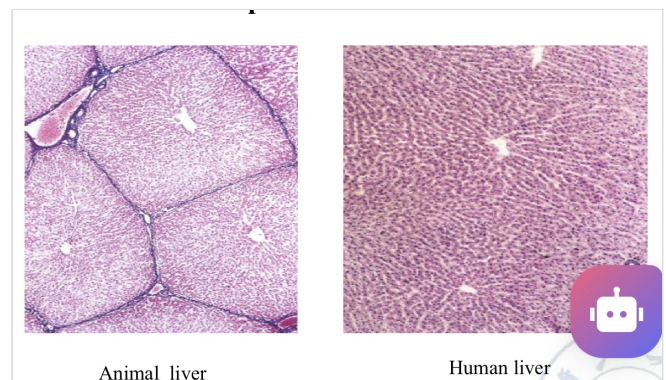


Fig 1.3 Animal liver (left) versus human liver (right) at low power. In the pig the lobules are outlined by obvious connective-tissue septa; in the human they are **not**. In a human slide you find the lobule by locating the central vein and the portal triads, not by looking for a boundary.

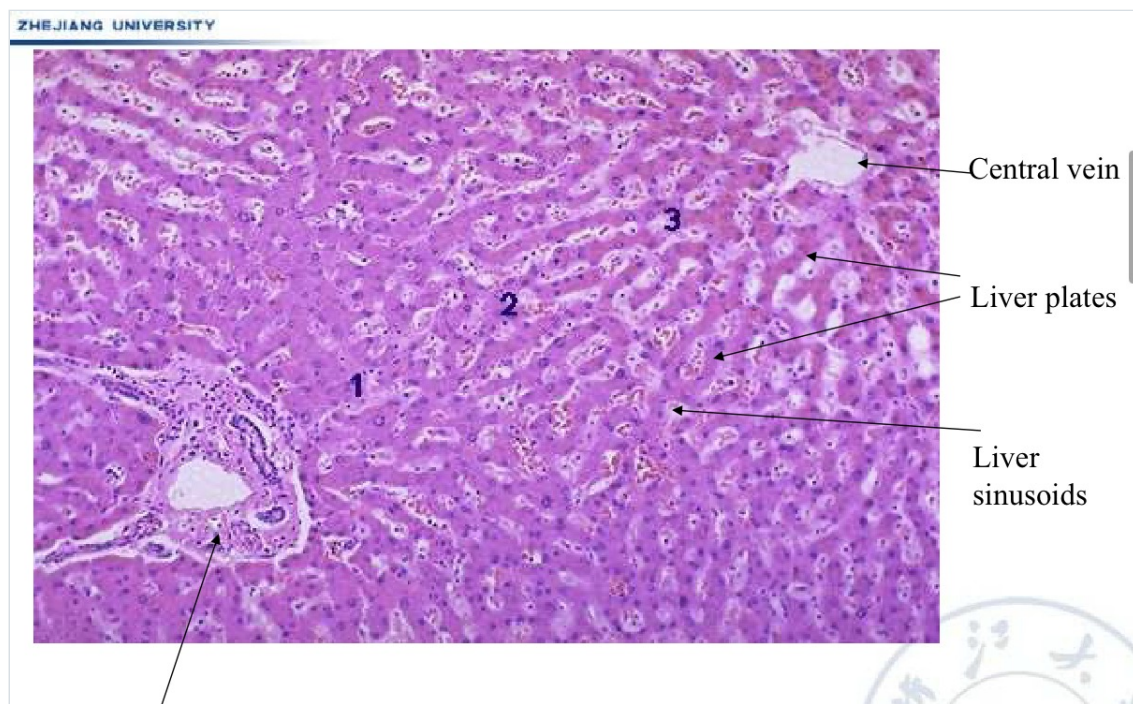


Fig 1.4 Normal human liver, labelled. **Central vein** (top right), **liver plates** radiating from it, **liver sinusoids** between the plates, and the **portal space** at lower left containing the interlobular bile duct, vein and artery. Fix these four names now — every slide in Part 4 is described in terms of them.

TABLE 1.2 THE FOUR STRUCTURES YOU MUST FIND ON ANY LIVER SLIDE

Structure	What it looks like	What happens to it in this session
Central vein	A single thin-walled vessel with no companion artery or duct, at the centre of the lobule	Your landmark for zone 3 — the centrilobular zone, furthest from the blood supply and therefore the first to show fatty change
Liver plates (cords)	One- to two-cell-thick rows of hepatocytes radiating from the central vein	In swelling and in fatty change the plates thicken and lose their radial order
Sinusoids	Wide, irregular capillaries between the plates, lined by endothelium and Kupffer cells	Compressed and narrowed whenever hepatocytes enlarge. This is the single most reliable low-power sign of hepatocyte swelling
Portal space (triad)	A connective-tissue island containing three structures: interlobular bile duct, vein and artery	Marks zone 1 — periportal , best oxygenated, last to be damaged by hypoxia

BENCH TECHNIQUE — HOW TO TELL A CENTRAL VEIN FROM A PORTAL VEIN

Count the companions. A central vein travels *alone* and hepatocyte plates run straight up to its wall. A portal vein sits in a patch of connective tissue with *two* companions — a bile duct (cuboidal epithelium, small round lumen) and an artery (thick muscular wall, small lumen). If you can see a duct, you are in a portal tract. Students who confuse the two describe centrilobular lesions as periportal and lose the zone, which in liver pathology is half the diagnosis.

1.3 Stomach and Intestine

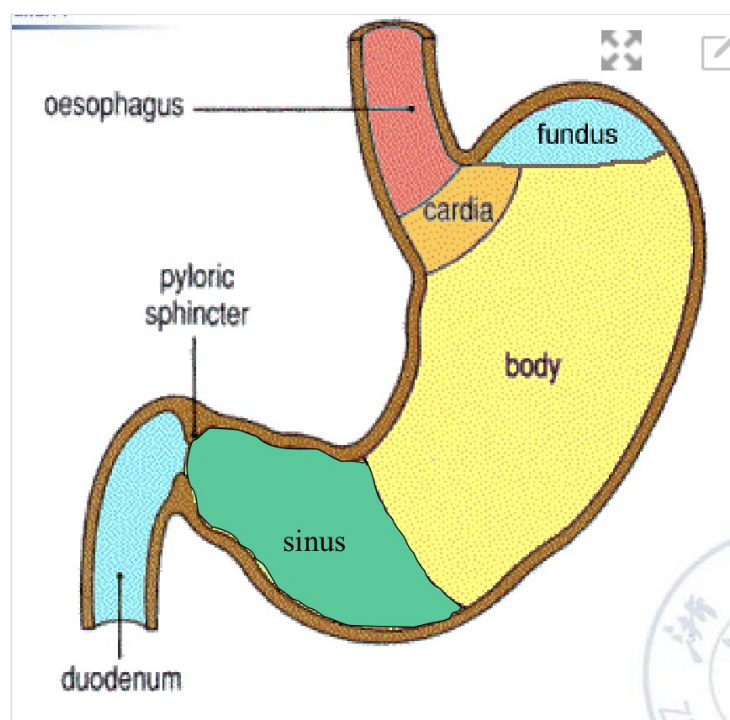


Fig 1.6 The five regions, in the order food meets them: **cardia → fundus → body → sinus (antrum) → pyloric sphincter → duodenum**. Naming the region is part of describing any gastric specimen, because several diseases have strong regional preferences.

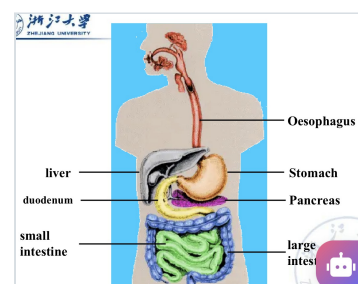


Fig 1.5 The tract as a whole. **Site:** abdominal cavity. **Function:** to supply nutritive substance — fat, glycogen and protein, which provide energy — and to evacuate the residue and toxins.

The four layers — the same everywhere

From the oesophagus to the rectum the wall has the same four layers in the same order. Learn them once:

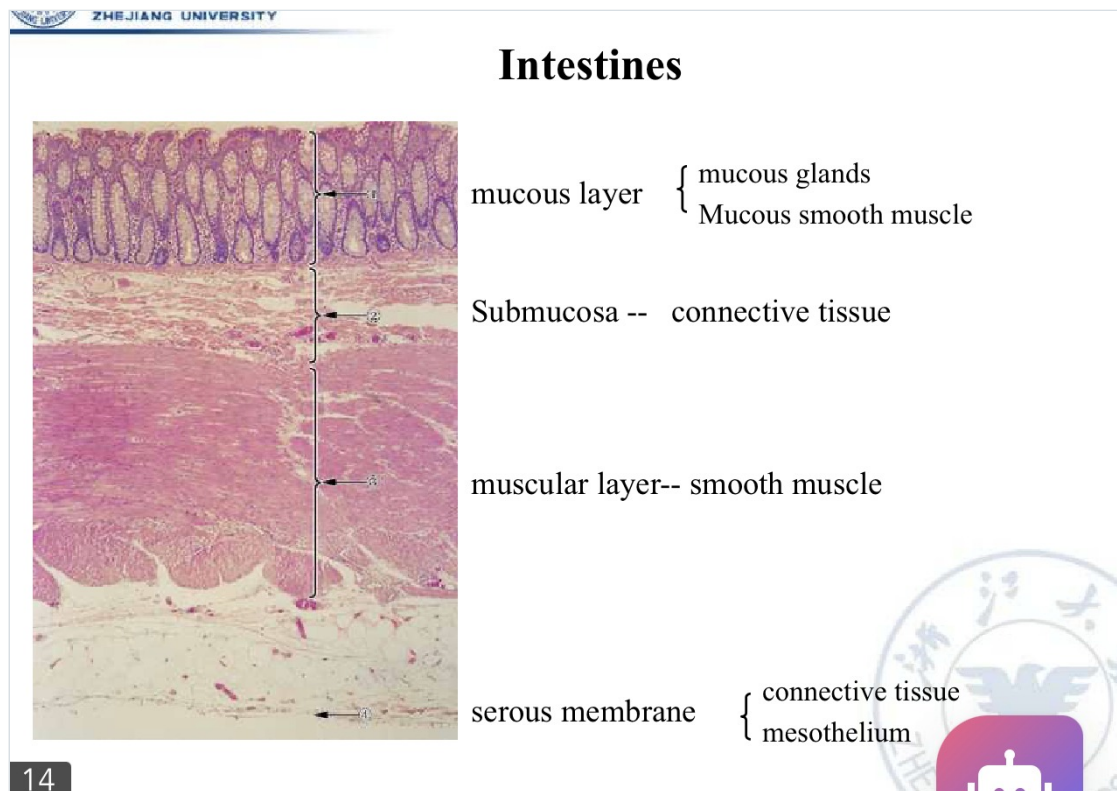


Fig 1.7 Intestinal wall, all four layers in one field. **1** mucous layer (mucous glands plus muscularis mucosae) · **2** submucosa (connective tissue) · **3** muscular layer (smooth muscle) · **4** serous membrane (connective tissue plus mesothelium). The muscular layer is the thick pink band — use it to orient yourself, then work outwards and inwards from there.



Fig 1.8 Normal small intestinal mucosa: **long villi with occasional goblet cells**. The villi exist to provide a large surface area for digestion and absorption — which is why villous shortening or flattening, in any disease, translates directly into malabsorption.

TABLE 1.3 THE FOUR LAYERS AND WHAT EACH CONTAINS

Layer	Contains
Mucous layer (mucosa)	Mucous glands and the muscularis mucosae (a thin layer of smooth muscle). The functional surface — epithelium, lamina propria, muscularis mucosae
Submucosa	Connective tissue — plus the vessels, lymphatics and Meissner's plexus that supply the mucosa above it
Muscular layer	Smooth muscle — inner circular and outer longitudinal, with Auerbach's plexus between them
Serous membrane (serosa)	Connective tissue plus mesothelium — a single flat cell layer. The same mesothelium that lines the pleura, which matters in §3.4

EXAM TRAP — STOMACH OR INTESTINE?

Both have the same four layers, so the layers do not tell you which. **Look at the mucosal surface.** The stomach has a flat surface pitted by **gastric pits** leading down to tubular glands; the small intestine has **villi projecting upwards** with crypts between them; the colon has **neither villi nor pits** — just straight test-tube glands packed with goblet cells. Surface shape decides it in a second at 4×.

PART TWO

02

Reversible Injury — the Five Changes

Water, fat, protein, mucin, calcium. Five substances, five places they should not be, five appearances. Learn what accumulates and why the cell cannot clear it, and the morphology follows without being memorised separately.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Explain why a failing sodium pump makes a cell swell.
- Give four mechanisms by which fat accumulates in a hepatocyte.
- Name the three kinds of hyaline degeneration and where each occurs.
- Separate dystrophic from metastatic calcification by serum calcium.

2.1 Where This Session Sits on the Injury Spectrum

1
Adaptation — atrophy, hypertrophy, hyperplasia, metaplasia. A new stable state. Session 1.
2
Reversible injury — the cell is overwhelmed but the damage can still be undone. Swelling and fatty change. This session.
3
Irreversible injury — the point of no return has passed. Necrosis or apoptosis. A later session.

The five changes in this session are not all injuries to a single cell, and it is worth being precise about that from the start:

TABLE 2.1 THE FIVE CHANGES AT A GLANCE

Change	What accumulates	Where	Gross appearance
Cellular swelling (cloudy swelling, hydropic change)	Water	Inside the cell — cytosol and organelles	Organ enlarged, pale, dull , capsule tense, edges rounded
Fatty change (steatosis)	Triglyceride	Inside the cell — cytoplasmic vacuoles	Organ enlarged, yellow, greasy, soft
Hyaline degeneration	Protein	Connective tissue, vessel wall, or inside the cell	Grey-white, translucent, frosted glass , dense and uniform
Mucoid change (myxoid)	Mucin — mucopolysaccharide	Connective tissue matrix	Grey-blue, gelatinous, translucent , slippery
Pathological calcification	Calcium salts	Dead tissue (dystrophic) or normal tissue (metastatic)	Grey-white, gritty, hard as gravel , sharply bounded

HIGH-YIELD — THE TWO COLOUR RULES

Almost every gross specimen in this session can be sorted by colour before you know anything else.

Pale and dull → water (cellular swelling). **Yellow and greasy** → fat. **Grey-white and translucent** → hyaline. **Grey-blue and gelatinous** → mucin. **Grey-white and gritty** → calcium.

The two grey-whites are told apart by *touch*, not sight: hyaline is smooth and firm, calcium is hard and gritty. That is why the specimen descriptions keep mentioning texture.

2.2 Cellular Swelling — Water

DEFINITION

Cellular swelling — also called **cloudy swelling** (gross) and **hydropic degeneration** or **hydropic change** (microscopic) — is the accumulation of **water inside the cell** because energy-dependent ion transport has failed. It is the **first manifestation of almost all forms of injury** to a cell, and it is fully reversible.

○ MECHANISM — FROM A FAILING PUMP TO A SWOLLEN ORGAN

- 1 An insult reduces **ATP production** — most often **hypoxia or ischaemia**, but also toxins, infection or free radicals.
WHY: Every subsequent step follows from energy failure alone. This is why swelling is non-specific: any injury that costs the cell ATP produces it.
- 2 The **Na⁺/K⁺-ATPase** of the cell membrane slows or stops.
WHY: This pump consumes a large share of a cell's total ATP budget — in the renal proximal tubule, the majority of it. It is the first thing to fail and the most consequential.
- 3 **Sodium accumulates inside the cell** and potassium leaks out.
WHY: The pump normally exports three sodium ions for every two potassium imported, so stopping it is not neutral — it leaves a net osmotic load inside.
- 4 **Water follows sodium in**, to restore osmotic balance.
WHY: The membrane is freely permeable to water. Solute cannot leave, so water must enter — there is no alternative outcome.
- 5 The cell and its organelles swell. The **endoplasmic reticulum dilates**, ribosomes detach, and mitochondria swell.
*WHY: Dilated ER and swollen mitochondria scattered through the cytoplasm are what the eye reads as **fine granularity** — the “cloudy” of cloudy swelling. At a further stage the dilated ER coalesces into clear spaces and the cell looks **reticulate**, then **transparent**.*
- 6 In the most severe reversible stage, the cell becomes a **balloon** — hugely distended and almost clear.
*WHY: Even here the nucleus is **normal**. That is the whole distinction from necrosis: a swollen cell with an intact nucleus is alive and salvageable; nuclear pyknosis, karyorrhexis or karyolysis means it is not.*
- 7 Multiplied over the whole organ, the swollen cells **increase its volume and mass**.
*WHY: The organ enlarges, its capsule becomes tense, its edges round off, and the cut surface bulges. It goes **pale** because the swollen cells compress the capillaries and squeeze blood out, and **dull** because the cytoplasm has lost its normal optical clarity.*

NET RESULT An enlarged, pale, dull, tense organ whose cells are swollen and granular but whose nuclei are entirely normal. Restore the oxygen and every step reverses.

HIGH-YIELD — WHY THE PROXIMAL TUBULE, AND WHY THE LIVER

The lecture makes a point of this. Hydropic change can occur in any renal tubule, **but the proximal tubule is by far the commonest site**, and the reason is metabolic: it is the most metabolically active segment, reabsorbing most of the water, sodium, glucose and amino acids of the filtrate. That work demands a great deal of energy and a great many mitochondria — which makes the cell exquisitely sensitive to **hypoxia, poisoning and infection**. When energy metabolism fails, its sodium pump fails first. The same logic explains the liver: the organ with the highest metabolic turnover shows metabolic injury first.

2.3 Fatty Change — Triglyceride

DEFINITION

Fatty change (steatosis, fatty degeneration) is the abnormal accumulation of **triglyceride within parenchymal cells**. It is most often seen in the **liver** — the organ central to lipid metabolism — and also in the heart, muscle and kidney.

Normal hepatocytes receive free fatty acids, esterify them to triglyceride, package the triglyceride with **apo-protein** into very-low-density lipoprotein, and export it. Fat accumulates when any step of that pipeline is disrupted — and there are four places it can break:

TABLE 2.2 FOUR WAYS TO MAKE A FATTY LIVER

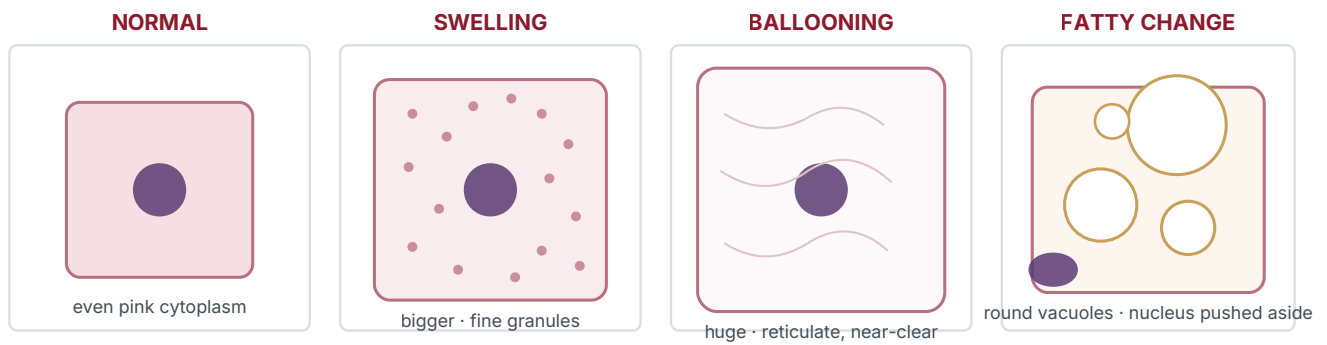
Mechanism	Classic cause	Why it works
Increased delivery of free fatty acids to the liver	Starvation, diabetes mellitus, obesity	Peripheral fat is mobilised faster than the liver can process it
Increased synthesis of fatty acid	Alcohol	Ethanol metabolism raises NADH, which pushes the cell towards lipid synthesis
Decreased oxidation of fatty acid	Hypoxia , alcohol, toxins	β -oxidation is mitochondrial and oxygen-dependent; fat that is not burned is stored
Decreased apoprotein synthesis → impaired export	Carbon tetrachloride , protein malnutrition (kwashiorkor)	Triglyceride cannot leave the cell without its protein carrier, so it stays

HIGH-YIELD — WHY ZONE 3 GOES FIRST

Blood enters the lobule at the portal triads and leaves at the central vein, so hepatocytes around the **central vein (zone 3, centrilobular)** receive blood that has already been stripped of oxygen by the cells upstream. In any hypoxic cause of fatty change — cardiac failure, anaemia, shock — **zone 3 is affected first and most severely**. Naming the zone in your description tells the reader the mechanism, which is exactly what a pathologist is paid to do.

The distinction the lecturer asks for

Both cellular swelling and fatty change produce a large, soft, pale-ish organ full of cells with clear-looking cytoplasm. This diagram is the answer to the difference, and it is worth studying before you touch the slide:



The deciding feature

Water spreads through the whole cytoplasm and leaves the nucleus central. Fat forms discrete round vacuoles of differing size, and a large one pushes the nucleus to the cell edge.

Fig 2.1 Hepatocyte appearances along the reversible-injury spectrum. Note that only the fourth panel has a displaced nucleus — that is the single most reliable discriminator at 40×.

BENCH TECHNIQUE — FAT IS NOT THERE ON YOUR SLIDE

Routine processing uses organic solvents that **dissolve lipid away**, so on an H&E slide the fat has gone and you are looking at the empty holes it left. That is why the vacuoles are perfectly clear and sharply outlined. To show fat you must avoid the solvents: cut a **frozen section** and stain with **Sudan III**, **Sudan IV** or **oil red O**, which turn lipid orange-red. The gross specimen in §3.2 was tested exactly this way and stained orange-yellow.

2.4 Hyaline Degeneration — Protein

DEFINITION

Hyaline degeneration (hyalinization) is not a single disease but a **descriptive term**: any accumulation that appears **homogeneous, glassy, translucent and strongly eosinophilic** on H&E. From Greek *hyalos*, glass. The underlying material is almost always **protein**, but a different protein in each setting.

TABLE 2.3 THE THREE KINDS OF HYALINE DEGENERATION

Kind	Where	What the material is, and the classic example
Connective tissue hyaline	Dense fibrous tissue — old scars, thickened serous membranes	Collagen bundles fuse and lose their individual outlines. This session: the thickened pleura (§3.4) ; also keloid and the hyalinised scar of an old healed lesion
Vascular hyaline (hyaline arteriosclerosis)	Walls of arterioles and small arteries	Plasma proteins leak into the wall and basement-membrane material accumulates, narrowing the lumen. Benign hypertension and diabetes mellitus. This session: Slide 43 (§4.4)
Intracellular hyaline	Inside individual cells	Protein droplets: hyaline droplets in proximal tubular cells in proteinuria; Russell bodies (immunoglobulin) in plasma cells; Mallory bodies (keratin) in alcoholic hepatitis

EXAM TRAP — HYALINE IS A LOOK, NOT A SUBSTANCE

The commonest error is to treat “hyaline” as one material. It is not. Collagen fused in a scar, plasma protein in an arteriolar wall, immunoglobulin in a plasma cell and keratin in a hepatocyte are four chemically unrelated substances that happen to share one microscopic appearance: **homogeneous, glassy and pink**. If a question asks “what is hyaline made of?”, the only correct answer begins “*it depends where it is —*”.

2.5 Muroid Change — Mucin

DEFINITION

Muroid change (mucinous or myxoid degeneration) is the accumulation of **mucin — a mucopoly-saccharide-protein complex — in connective tissue** where it does not normally occur in that quantity. The tissue becomes **grey-blue, translucent, gelatinous and slippery**.

Mucin is a normal product of epithelium — goblet cells make it by the litre. Muroid *degeneration* is different: the mucin is in the **connective tissue stroma**, secreted by fibroblasts and mesenchymal cells rather than by glands. It is seen in:

- **Tumours of mesenchymal origin** — the myxoid areas of a fibroadenoma (this session’s specimen 11), myxomas, some sarcomas.
- **Hypothyroidism** — myxoedema, where mucin accumulates in the dermis and gives the characteristic non-pitting puffiness.
- **Connective-tissue diseases and malnutrition**, where ground substance accumulates as the matrix loosens.

BENCH TECHNIQUE — PROVING IT IS MUCIN

On H&E mucin is pale, blue-grey and stringy, and can be mistaken for oedema or for a processing artefact. Two special stains settle it: **Alcian blue** (acid mucins turn blue) and **mucicarmine** (mucin turns red). Both are quick, and the question “is this mucin or just fluid?” is exactly the sort a good description flags for the reader.

2.6 Pathological Calcification — Calcium

DEFINITION

Pathological calcification is the abnormal deposition of **calcium salts**, together with smaller amounts of iron, magnesium and other minerals, in tissue other than bone and teeth. It comes in exactly two forms, and telling them apart is a matter of one blood test.

TABLE 2.4 DYSTROPHIC VERSUS METASTATIC CALCIFICATION

	Dystrophic	Metastatic
Tissue	Dead or dying — necrosis, old thrombus, damaged valve, atheromatous plaque	Normal, living tissue
Serum calcium	Normal	Raised — hypercalcaemia
Underlying problem	Local — the tissue is abnormal	Systemic — calcium metabolism is deranged
Typical causes	Tuberculous caseous necrosis · atherosclerotic plaque · old infarct · calcified valve	Hyperparathyroidism · bone destruction by tumour or myeloma · vitamin D excess · renal failure
Favoured sites	Wherever the damage happens to be	Acid-secreting tissues: gastric mucosa, kidney, lung, systemic arteries and pulmonary veins
This session	Both specimens — the calcified tuberculous lesion of lung (\$3.6) and the coronary plaque (\$4.5)	Not represented

○ MECHANISM — WHY DEAD TISSUE ATTRACTS CALCIUM

- 1 A cell dies. Its **membrane pumps stop**, including the calcium pumps that normally keep intracellular calcium a thousandfold lower than outside.
WHY: A living cell spends energy continuously to keep calcium out. Death removes that defence.
- 2 Calcium floods into the dead cell and its mitochondria, and **membrane phospholipids** become exposed, binding it.
WHY: Phospholipid head groups are negatively charged and bind calcium avidly. Dead membranes are therefore nucleation sites — the process starts wherever membranes have broken.
- 3 Phosphate, generated locally by phosphatases released from the dying cells, joins the calcium to form the first **crystal nucleus**.
WHY: This is the initiation step. It is local and requires no systemic abnormality, which is why serum calcium is normal.
- 4 The nucleus grows by **propagation**, adding calcium and phosphate until it becomes crystalline hydroxyapatite.
WHY: Growth is self-sustaining once started, which is why calcification is permanent — the body has no mechanism to remove it.

NET RESULT A permanent, grey-white, gritty deposit that is **hard as gravel** under the knife and that appears **deep purple-blue** on H&E — because haematoxylin is a basic dye and calcium salts are strongly basophilic.

HIGH-YIELD — PURPLE-BLUE ON A PINK SLIDE

On an H&E section almost everything is pink (eosinophilic) except nuclei. A **deep purple-blue amorphous deposit that is not a nucleus** — irregular, granular, in a place where no cell should be — is calcium until proved otherwise. In \$4.5 you will find exactly that in the necrotic centre of a coronary plaque.

PART THREE

03

Gross Specimens

Seven specimens, one card each: what you see, why it looks that way, what to call it, and the sentence to write. Cover everything below *What you see* and produce the rest yourself — that is the bench task and the exam task, unchanged.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Describe and diagnose all seven gross specimens unaided.
- Separate the tigroid heart from the fat heart on sight.
- Explain why a cloudy-swollen liver is pale and a fatty liver is yellow.

3.1 No. 5 — Cloudy Swelling of the Liver

GROSS SPECIMEN NO. 5 · GROSS · DEMONSTRATION (示教)

Cloudy swelling of the liver

WHAT YOU SEE

- The **liver volume increases** and the **capsule is tense**.
- The **liver margin is noticeably dull and round** — the normal sharp edge has gone.
- On the cut surface, the **capsule at the edge is inverted**, the **parenchyma is significantly elevated**, and the **portal tissue is relatively sunken**.
- The parenchyma is **pale in colour and has lost its normal lustre, resembling boiling water**.

WHY IT LOOKS LIKE THAT

Every feature is a mechanical consequence of hepatocytes taking on water (§2.2).

Because: Billions of cells each a little larger add up to a **larger organ**; the capsule, which cannot stretch to match, becomes **tense**, and the sharp edge is pushed outwards into a **blunt curve**. When the organ is cut, the swollen parenchyma is under pressure and **bulges out** of the plane of section, dragging the cut edge of the capsule over with it (the “inverted” capsule) — while the portal tracts, which are fibrous and do not swell, stay behind and therefore look **sunken**. The **pallor** is compression of the sinusoids by the swollen cells, squeezing blood out of the tissue. The **loss of lustre** is the granular, dilated organelles scattering light instead of transmitting it — which is exactly what makes a poached egg white opaque, and exactly what the phrase “resembling boiling water” is describing.



Fig 3.1 Compare the edge and the colour with Fig 1.1. The margin is blunt and rounded, and the parenchyma is uniformly pale and matt.

PATHOLOGICAL DIAGNOSIS

Cloudy swelling of the liver (cellular swelling / hydropic change)

WRITE IT LIKE THIS

Organ: Liver. **Description:** The liver is increased in volume and the capsule is tense. The inferior margin is blunt and rounded rather than sharp. On the cut surface the capsular edge is inverted, the parenchyma bulges above the plane of section, and the portal tracts are relatively depressed. The parenchyma is diffusely pale and has lost its normal lustre. No nodularity, fibrosis or focal lesion is seen. **Diagnosis:** Cloudy swelling of the liver.

3.2 No. 6 — Fatty Degeneration of the Liver

GROSS SPECIMEN NO. 6 · GROSS · HOMEWORK

Fatty degeneration of the liver (hepatic steatosis)

WHAT YOU SEE

- The **liver is enlarged**, the **edges are obtuse** (blunt and round), and the **capsule is tight**.
- The **surface and the cut section are both yellow** — described on the specimen card as **grayish yellow**.
- The **cut surface has a greasy feeling** to the touch.
- The **texture is soft and light**.
- Stained with **Sudan III**, the liver turned **orange-yellow** — **positive for fat**.

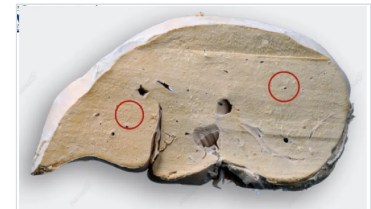


Fig 3.2 The cut surface is uniformly yellow. Put this beside Fig 3.1: same enlarged organ, same blunt edge, entirely different colour — and that colour is the diagnosis.

WHY IT LOOKS LIKE THAT

Triglyceride has accumulated in the hepatocytes faster than the liver can export it (§2.3).

Because: Fat is **yellow**, so enough of it inside enough cells colours the whole organ. Fat is **less dense than protein and water**, so the organ becomes light for its size — an enlarged liver that feels unexpectedly light is a real physical finding, not a figure of speech. Fat is **greasy**, so the cut surface leaves a film on the blade and on the glove. And fat is **soft**, so the organ loses its normal firmness. The enlargement and blunt edges are shared with cloudy swelling, because both are cells taking on volume; it is the **colour and the greasiness** that separate them. The Sudan III result is the confirmation: a fat-soluble dye partitions into the lipid and stains it orange, which is only possible if the vacuoles really do contain fat.

PATHOLOGICAL DIAGNOSIS

Fatty degeneration of the liver (hepatic steatosis)

WRITE IT LIKE THIS

Organ: Liver. **Description:** The liver is enlarged with obtuse, rounded edges and a tight capsule. Both the external surface and the cut surface are diffusely greyish-yellow. The cut surface is greasy to the touch and the parenchyma is soft, and the organ is light for its size. Sudan III staining gives an orange-yellow colour, positive for fat. No nodules, fibrous bands or focal lesions are seen. **Diagnosis:** Fatty degeneration of the liver.

CLINICAL CORRELATION — WHO HAS A FATTY LIVER

The two commonest causes in practice are **alcohol** and **metabolic syndrome** (obesity, type 2 diabetes, insulin resistance — non-alcoholic fatty liver disease). Fatty change by itself is **reversible and asymptomatic**: stop the alcohol, lose the weight, and the liver returns to normal over weeks. What makes it matter is what comes next — if the insult continues, steatosis progresses through steatohepatitis to fibrosis and cirrhosis, and cirrhosis is not reversible. The specimen in front of you is the last easily-fixable stage.

3.3 No. 7 and No. 8 — Tigroid Heart and Fat Heart

These two are deliberately set side by side, and the lecturer names *identifying* them as an objective in its own right. They sound similar and they are entirely different lesions: in one the fat is **inside the muscle cells**, in the other it is **outside them**.

GROSS SPECIMEN NO. 7 · GROSS

Fatty degeneration of the myocardium — the tigroid heart

WHAT YOU SEE

- Under the **endocardium and papillary muscle of the left ventricle** there are **many orange-red stripes** on a **light brown background**.
- The pattern **resembles tiger skin** — alternating pale and darker bands.
- The change is **beneath the endocardium**, not on the outer surface.

WHY IT LOOKS LIKE THAT

The stripes are **a map of the oxygen supply**.

Because: Fatty change in myocardium follows **hypoxia** — classically severe anaemia, or chronic heart failure. Oxygen reaches myocytes from the coronary capillaries, so the cells immediately **around a capillary** stay well oxygenated and metabolise their fatty acids normally, while the cells **furthest from one** run short, cannot complete β -oxidation, and accumulate triglyceride. Because the capillaries run in parallel bundles, the affected and unaffected zones alternate in **parallel bands** — and pale yellow fat-laden muscle next to normal red-brown muscle is exactly a tiger stripe. The subendocardium is worst affected because it is the furthest territory from the epicardial coronary arteries and works under the highest wall stress.

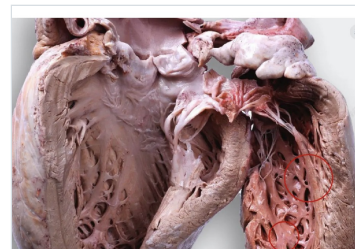


Fig 3.3 Left ventricle opened. The circled areas show the orange-red streaks on a light brown background — the tiger-skin pattern that names the lesion.

PATHOLOGICAL DIAGNOSIS

Fatty degeneration of the myocardium — tigroid heart

WRITE IT LIKE THIS

Organ: Heart. **Description:** The left ventricle has been opened. Beneath the endocardium and over the papillary muscles there are numerous fine orange-red stripes alternating with a light brown background, producing a banded, tiger-skin pattern. The ventricular wall is not thickened and no scar or infarct is seen. **Diagnosis:** Fatty degeneration of the myocardium (tigroid heart).

GROSS SPECIMEN NO. 8 · GROSS

Fat heart (fatty infiltration of the heart)

WHAT YOU SEE

- The amount of adipose tissue is increased, on both the **surface and the cut surface**.
- The excess fat is in the **pericardium, especially over the cardiac apex**.
- **Fat has infiltrated between the cardiac muscle fibres**, extending inwards from the epicardial surface.

WHY IT LOOKS LIKE THAT

This is not degeneration at all — it is **adipose tissue growing where it already lived**.

Because: The normal epicardium contains fat, particularly along the atrioventricular and interventricular grooves and over the apex. In obesity — and to a lesser degree in ageing — that existing fat simply increases in bulk and begins to extend inwards along the connective-tissue planes between muscle bundles. The myocytes themselves are **completely normal**: nothing has accumulated inside them. That is the whole difference from specimen 7, and it is the difference between a **metabolic injury to muscle cells** and an **excess of a normal tissue in a normal place**. In extreme cases the infiltrating fat separates and atrophies the muscle bundles it surrounds, and the wall can become weak — but the fat is still outside the cells.

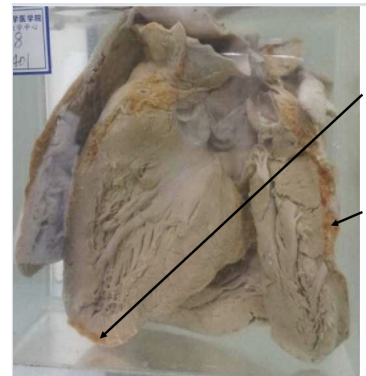


Fig 3.4 The fat is **on the outside**, thickest over the apex, and on the cut surface it can be traced running down between the muscle bundles.

PATHOLOGICAL DIAGNOSIS

Fat heart — fatty infiltration of the myocardium

WRITE IT LIKE THIS

Organ: Heart. **Description:** There is a marked increase in epicardial adipose tissue, most pronounced over the cardiac apex, visible both on the external surface and on the cut surface. Strands of adipose tissue extend inwards from the epicardium between the myocardial fibres. The myocardium itself is of normal colour and the ventricular wall is not thinned. **Diagnosis:** Fat heart (fatty infiltration of the myocardium).

HIGH-YIELD — INSIDE VERSUS OUTSIDE, IN ONE LINE

Tigroid heart = fat INSIDE the myocytes (fatty degeneration, a metabolic injury, caused by hypoxia — think severe anaemia). **Fat heart = fat OUTSIDE the myocytes** (fatty infiltration, not an injury to the cells at all — think obesity). One is a disease of the muscle cell; the other is a disease of the patient's waistline. Both are asked, and they are asked together.

3.4 No. 10 — Hyaline Degeneration of the Thickened Pleura

GROSS SPECIMEN NO. 10 · GROSS

Hyaline degeneration of the thickened pleura

WHAT YOU SEE

- Section of lung (thoracic) tissue showing **multiple tuberculous cavities**, with **caseous necrosis attached to the inner wall** of the cavities.
- The lung is **collapsed and atrophic**.
- The **visceral pleura is adherent to the parietal layer** and is **markedly thickened**.
- The thickened pleura is **greyish white, translucent like frosted glass**, and of **uniform, dense texture**.

WHY IT LOOKS LIKE THAT

This is the end-stage of a long-healed inflammation, and every feature is a step in that story.

Because: *Chronic tuberculous inflammation of the lung reached the pleural surface. The pleura responded with **fibrosis** — fibroblasts laying down collagen — and the inflamed visceral and parietal layers stuck together. Over years, the collagen bundles in that scar **fused and lost their individual outlines**, becoming a homogeneous mass: that is **connective-tissue hyaline degeneration** (§2.4). Homogeneity is what makes it **translucent** — light passes through a uniform material but scatters at every boundary in a fibrous one, which is exactly why frosted glass is the right comparison. The **collapse and atrophy** of the underlying lung follow from the cavities destroying parenchyma and the rigid pleural rind preventing expansion.*

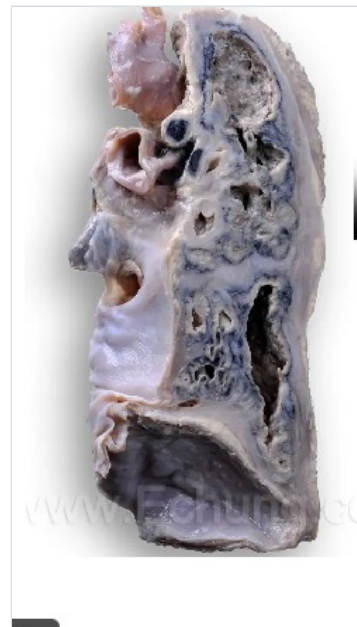


Fig 3.5 Cut section of lung. The pale, dense rind along the outer edge is the thickened, hyalinised pleura; the cavities and collapsed parenchyma lie within.

PATHOLOGICAL DIAGNOSIS

Hyaline degeneration of the thickened pleura, secondary to chronic pulmonary tuberculosis

WRITE IT LIKE THIS

Organ: Lung with pleura. **Description:** The cut surface shows multiple cavities with caseous necrotic material adherent to their inner walls, and the intervening parenchyma is collapsed and atrophic. The visceral pleura is adherent to the parietal pleura and is markedly thickened, forming a greyish-white, translucent rind with a uniform, dense, frosted-glass appearance. **Diagnosis:** Hyaline degeneration of the thickened pleura, in chronic pulmonary tuberculosis.

3.5 No. 11 — Mucoïd Degeneration of Fibroadenoma

GROSS SPECIMEN NO. 11 · GROSS

Mucoïd (myxoïd) degeneration of a fibroadenoma

WHAT YOU SEE

- A **subcutaneous nodular mass** with an **oval shape**.
- **Clear grey and white boundaries** and a **complete capsule**.
- On the cut surface, **translucent (refractile) grey and white fibre streaks**.

WHY IT LOOKS LIKE THAT

A fibroadenoma is a benign tumour of breast **stroma and glands**, and it is the stroma that degenerates.

Because: The tumour is well circumscribed with a complete capsule because it is **benign** — it pushes the surrounding tissue aside rather than invading it, and the compressed tissue forms the capsule. That single feature is worth stating in the description, because it is what separates a fibroadenoma from a carcinoma at the naked-eye level. Within the fibrous stroma, the mesenchymal cells accumulate **mucin** in the matrix, which becomes loose, gelatinous and grey-blue — **mucoïd degeneration** (§2.5). Elsewhere in the same nodule, older collagen has fused into dense white **hyaline** bands. Two degenerations, one specimen.

PATHOLOGICAL DIAGNOSIS

Fibroadenoma with mucoïd (myxoïd) degeneration

WRITE IT LIKE THIS

Organ: Breast. **Description:** An oval, well-circumscribed subcutaneous nodule with a complete capsule and clear grey-white boundaries. The cut surface shows translucent grey-white fibrous streaks together with irregular, softer grey-blue areas of gelatinous appearance. There is no invasion of surrounding tissue, no necrosis and no haemorrhage. **Diagnosis:** Fibroadenoma showing mucoïd (myxoïd) degeneration.



Fig 3.6 The cut surface of the nodule. Note the two different textures present together — see the Exam Trap below before you label either of them.

EXAM TRAP — THE LECTURER'S OWN CORRECTION

The slide for this specimen carries a red-boxed note from the lecturer flagging an **annotation error in the accompanying video**. The correction is:

- The **white stripe** is **hyaline degeneration** — dense, opaque, firm.
- The **irregular grey-blue area** is **mucoïd degeneration** — loose, translucent, gelatinous.

A lecturer who takes the trouble to correct his own material in a red box is telling you plainly which distinction he intends to examine. Learn it in this direction: **white and dense = hyaline · grey-blue and gelatinous = mucoïd**. Note also the honest lesson buried here — two degenerations can and do coexist in one lesion, so “which one is it?” is sometimes the wrong question and “where is each?” is the right one.

3.6 No. 12 — Calcified Lesion of the Lung

GROSS SPECIMEN NO. 12 · GROSS

Calcified lesion of the lung — healed tuberculosis

WHAT YOU SEE

- A **grey-white calcified lesion at the apex of the lung, near the pleura.**
- Its surface is **fine-grained** and it is **hard as gravel.**
- The **boundary is clear** — sharply demarcated from the surrounding lung.
- It is **surrounded by black carbon** (anthracotic pigment).

WHY IT LOOKS LIKE THAT

This is a tuberculous focus that the immune system won, decades ago.

Because: Primary tuberculosis typically seeds the **apex** — the best-ventilated, most oxygen-rich part of the lung, which suits an obligate aerobe like *Mycobacterium tuberculosis*. The lesion caseated, the host walled it off with fibrous tissue, and the dead caseous material remained. Dead tissue cannot pump calcium out (§2.6), so calcium salts nucleated on its broken membranes and grew until the whole focus became stone. The **sharp boundary** is the fibrous wall the host built. The **gritty hardness** is crystalline hydroxyapatite — you can feel the knife catch. The **black surround** is unrelated: it is inhaled carbon, normal in any adult city-dweller, and it is worth saying so in the description rather than leaving the reader to wonder whether it is part of the lesion.



Fig 3.7 The pale, sharply bounded focus near the apex, set in heavily anthracotic lung.

PATHOLOGICAL DIAGNOSIS

Dystrophic calcification in a healed tuberculous lesion of the lung

WRITE IT LIKE THIS

Organ: Lung. **Description:** At the apex, close to the pleura, there is a well-demarcated grey-white focus with a fine-grained surface which is stony hard to the knife. The lesion is sharply bounded from the surrounding parenchyma and is surrounded by black anthracotic pigment. No cavitation, consolidation or mass lesion is present elsewhere in the section. **Diagnosis:** Calcified (healed) tuberculous lesion of the lung — dystrophic calcification.

PART FOUR

04 Microscopic Slides

Five slides. The first three are the same lesion in two organs — swelling in kidney, swelling in liver, fat in liver — set up so you can see the difference for yourself. The last two are hyaline and calcium. Work each card at 4× first, exactly as you would at the bench.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Recognise hydropic change in a renal tubule and in a hepatocyte.
- Distinguish a hydropic vacuole from a fat vacuole at 40×.
- Identify hyaline change in a glomerulus and in an arteriolar wall.
- Recognise dystrophic calcification on an H&E section by its colour alone.

4.1 Slide 9 — Cloudy Swelling of the Kidney

SLIDE 9 · MICRO · H&E, HUMAN

Hydropic degeneration of renal tubular cells

WHAT YOU SEE

- The **epithelial cells of the renal proximal tubule are swollen**.
- The **cytoplasm is loose** and filled with a **large number of fine particles** and **amorphous vacuoles which are mildly eosin-stained**.
- **No significant differences could be found in the nuclei**.
- The tubular **lumina are narrowed** by the bulging cells.
- There is **interstitial vascular congestion** with a **small amount of inflammatory cell infiltration** in some areas.



Fig 4.1 Proximal tubules with swollen epithelium. The cells bulge into the lumen, which is narrowed to a slit; the nuclei are unremarkable.

WHY IT LOOKS LIKE THAT

The proximal tubule's sodium pump has failed and water has followed sodium in (§2.2).

Because: Two details in the description are doing real diagnostic work. First, the vacuoles are **amorphous** — shapeless and ill-defined, not round — because they are dilated cisterns of endoplasmic reticulum, which have no fixed shape. Second, they are **mildly eosin-stained** rather than water-clear, because they contain protein-rich cytosol, not pure water and certainly not empty space. Both features are the opposite of a fat vacuole, which is **round, sharply outlined and completely clear**. Third, the **nuclei are normal**, which is what tells you the cell is injured but alive. The narrowed lumina follow mechanically: the tubule wall cannot expand outwards against its basement membrane, so swelling cells can only expand inwards.

PATHOLOGICAL DIAGNOSIS

Cloudy swelling (hydropic degeneration) of the kidney

WRITE IT LIKE THIS

Tissue: Kidney. **Description:** The epithelial cells of the proximal convoluted tubules are diffusely swollen and bulge into the lumina, which are narrowed. Their cytoplasm is loose and contains numerous fine granules and ill-defined, faintly eosinophilic vacuoles. The nuclei show no significant change. There is interstitial vascular congestion with a scanty inflammatory cell infiltrate focally. Glomeruli are unremarkable and no necrosis is seen. **Diagnosis:** Cloudy swelling (hydropic degeneration) of the renal tubular epithelium.

4.2 Slide 46 — Hepatocellular Oedema

SLIDE 46 · MICRO · H&E, HUMAN

Hepatocellular oedema — cellular swelling of the liver

WHAT YOU SEE

- The **hepatocytes are diffusely swollen**.
- The **hepatic sinusoids are narrow or occlusive** — compressed by the enlarged cells.
- The **cytoplasm is loose, granular, reticulate, or transparent** — the four stages of hydropic degeneration, often all present on one slide.
- **A small number of hepatocytes are balloon-like** — hugely distended and nearly clear.

WHY IT LOOKS LIKE THAT

The same mechanism as Slide 9, in a different organ — and the four cytoplasmic appearances are a **time course**, not four different diseases.

Because: As water enters, the cytoplasm first becomes **loose**; swollen mitochondria and dilated ER then make it **granular**; as the dilated cisterns enlarge and run together they form a network of clear spaces, so the cell looks **reticulate**; finally the spaces coalesce completely and the cytoplasm looks **transparent** — the **balloon cell**. Seeing all four on one slide simply means you are looking at cells at different stages of the same process. The **narrowed sinusoids** are the most valuable low-power sign: normal liver looks airy because the sinusoids separate the plates, and a liver that looks solid and crowded at 4× has swollen hepatocytes even before you go to 40× to confirm it.

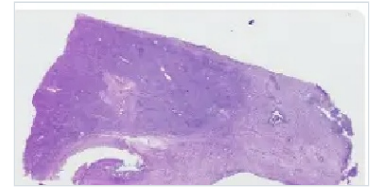


Fig 4.2 Low power. The tissue looks uniformly solid — the sinusoidal spaces that should separate the plates have been squeezed shut. That crowded, solid appearance is the low-power finding.

PATHOLOGICAL DIAGNOSIS

Hepatocellular oedema — cellular swelling (hydropic degeneration) of the liver

WRITE IT LIKE THIS

Tissue: Liver. **Description:** The lobular architecture is preserved. The hepatocytes are diffusely swollen and the intervening sinusoids are narrowed or obliterated. Hepatocyte cytoplasm is variably loose, finely granular, reticulate or transparent, and a small number of cells are markedly distended and almost clear (balloon cells). The nuclei are normally positioned and show no pyknosis or karyorrhexis. No fat vacuoles, necrosis or inflammatory infiltrate are seen. **Diagnosis:** Hepatocellular oedema (cellular swelling of the liver).

4.3 Slide 10 — Fatty Degeneration of the Liver

SLIDE 10 · MICRO · H&E, HUMAN · HOMEWORK

Fatty degeneration of hepatocytes

WHAT YOU SEE

- **Hepatocytes demonstrate diffuse fatty degeneration** — the change involves the whole section, not a focus.
- The **swollen hepatocytes contain abundant round vacuoles of varying sizes** in the cytoplasm.
- **A large vacuole can push the nucleus to one side** of the cell.
- The **hepatic sinusoids are compressed and narrowed**.

WHY IT LOOKS LIKE THAT

Triglyceride has accumulated as discrete droplets, and each descriptive word follows from the physics of a droplet.

Because: *Fat and cytosol do not mix, so the lipid forms separate droplets with a sharp interface — hence round and sharply outlined, unlike the shapeless dilated cisterns of hydropic change. Small droplets coalesce into larger ones over time, so a cell contains vacuoles of varying sizes — a single snapshot of an ongoing process. When one droplet becomes large enough it displaces everything else, including the nucleus, which ends up flattened against the cell membrane; a hepatocyte in that state looks exactly like an adipocyte and the appearance is called macrovesicular steatosis. And the fat is not actually in the section: the solvents of routine processing dissolved it, leaving the clean empty holes you are looking at (§2.3, Bench Technique).*

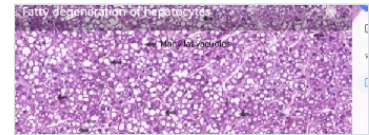


Fig 4.3 Diffuse fatty change. The vacuoles are round, sharply outlined and of clearly differing sizes — compare this texture with Fig 4.1, where the vacuoles have no edges at all.

PATHOLOGICAL DIAGNOSIS

Fatty degeneration (steatosis) of the liver

WRITE IT LIKE THIS

Tissue: Liver. **Description:** The lobular architecture is preserved. Hepatocytes throughout the section are enlarged and their cytoplasm contains numerous round, sharply defined, optically clear vacuoles of varying size. In many cells a single large vacuole displaces the nucleus to the periphery, giving a signet-ring appearance. The intervening sinusoids are compressed and narrowed. No inflammation, necrosis or fibrosis is identified. **Diagnosis:** Fatty degeneration (macrovesicular steatosis) of the liver.

HIGH-YIELD — THE FOUR-WORD TEST AT 40×

Put Slide 46 and Slide 10 side by side and ask four questions. **Shape?** Amorphous = water; round = fat. **Edge?** Ill-defined = water; sharp = fat. **Colour?** Faintly pink = water; completely clear = fat. **Nucleus?** Central = water; pushed to the edge = fat. Any one of the four will usually settle it; all four together always will.

4.4 Slide 43 — Hyaline Degeneration of Glomeruli and Arteriole

SLIDE 43 · MICRO · H&E, KIDNEY

Hyaline degeneration of glomeruli and arteriole

WHAT YOU SEE

- (2) **Glomeruli** converted into **solid, homogeneous, deeply eosinophilic pink balls** — the capillary loops and Bowman's space are no longer distinguishable.
- (1) **An arteriole** whose **wall is thickened by homogeneous pink material**, with the **lumen narrowed**.
- The **surrounding tubules** show atrophy and the interstitium some fibrosis.

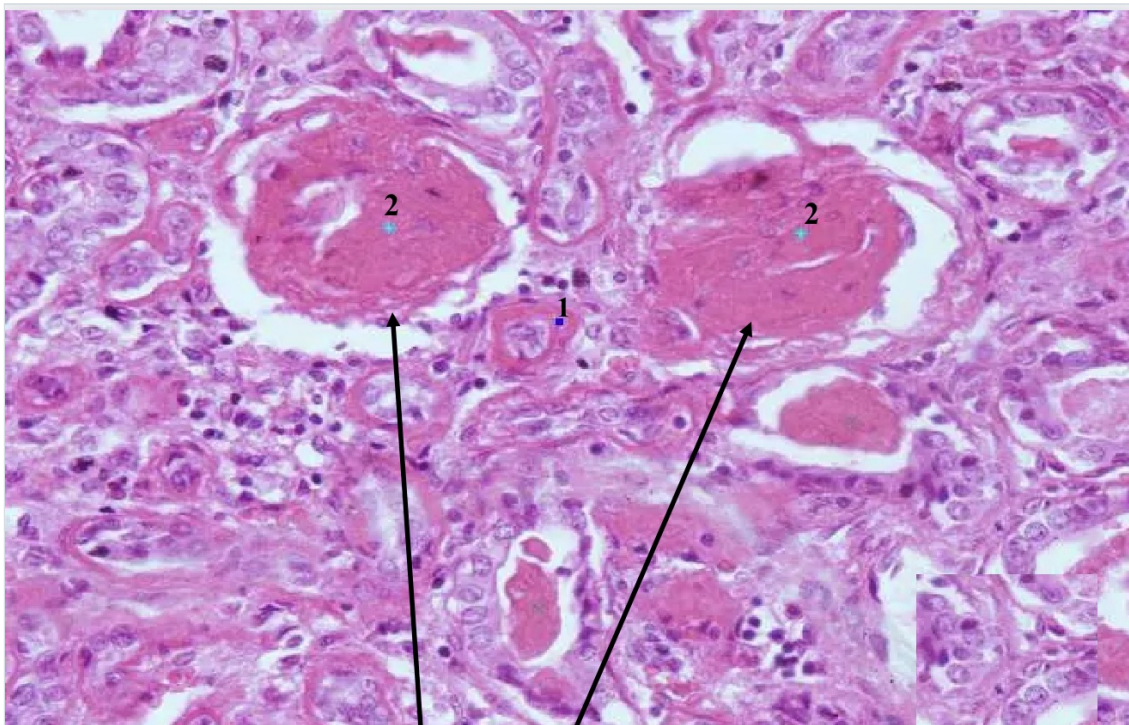


Fig 4.4 The two labelled lesions. The arrows point to the **hyalinised glomeruli (2)** — featureless pink spheres where a delicate capillary tuft should be — and **(1)** marks the thickened arteriole. Compare these glomeruli with the normal one in the Session 1 guide: there, you could count the capillary loops.

WHY IT LOOKS LIKE THAT

This is the kidney of long-standing **hypertension or diabetes**, and the two lesions are cause and effect.

Because: Chronic high pressure and endothelial injury make the arteriolar wall **leaky to plasma protein**. Protein seeps into the wall, and the smooth muscle cells respond by laying down excess basement-membrane material. Together these fill the wall with homogeneous pink protein — **hyaline arteriosclerosis** — and the lumen narrows. That narrowing is the cause of the second lesion: the glomerulus downstream is chronically **ischaemic**, its capillary tuft slowly collapses and is replaced by matrix and protein until the whole structure is a featureless ball. Why does it look **homogeneous**? Because the fine architecture that used to scatter light has been replaced by one uniform substance — the same reason the pleura in §3.4 looks like frosted glass. And the tubules atrophy because a glomerulus that no longer filters leaves its own nephron with nothing to do.

PATHOLOGICAL DIAGNOSIS

Hyaline degeneration of glomeruli and arterioles — benign nephrosclerosis

WRITE IT LIKE THIS

Tissue: Kidney. **Description:** Several glomeruli are converted into solid, homogeneous, strongly eosinophilic spheres in which capillary loops and Bowman's space can no longer be identified. An arteriole shows marked mural thickening by homogeneous eosinophilic material with narrowing of its lumen. The adjacent tubules are atrophic and the interstitium is mildly fibrotic. No cellular crescents, necrosis or inflammatory infiltrate are seen. **Diagnosis:** Hyaline degeneration of glomeruli and arterioles.

4.5 Calcified Lesion in the Coronary Artery

SLIDE · MICRO · H&E, CORONARY ARTERY

Coronary atherosclerosis with dystrophic calcification

WHAT YOU SEE

- **Atheromatous plaques formed in the intima** of the coronary artery, with **locally thickened walls and stenosis (grade IV)**.
- **Foam cells** in the **marginal area of the plaques**.
- In the **necrotic tissue at the centre** of the plaque, **scattered purple-blue granules or lumps** — dystrophic calcification.

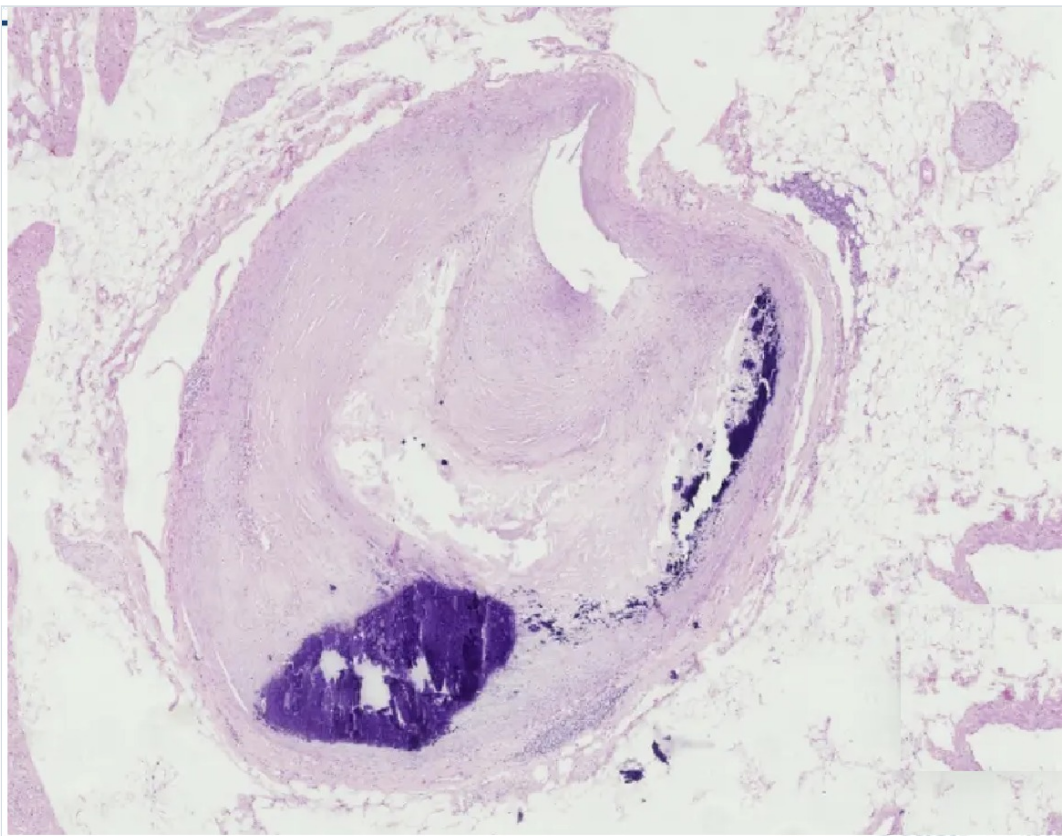


Fig 4.5 The plaque in cross-section. The lumen is reduced to a crescent at the upper left; the pale homogeneous mass is the fibrous cap and necrotic core; and the **dense purple-blue deposits** at lower and right are the calcification. Note that they sit within the necrotic core, exactly where §2.6 predicts.

WHY IT LOOKS LIKE THAT

Everything on this slide is a stage of one process, and the calcium is the last of them.

Because: *Lipid entered the injured intima; macrophages ate it and became **foam cells** — so called because the dissolved-out lipid leaves their cytoplasm frothy. Foam cells die, releasing their lipid and their enzymes, and the plaque acquires a **necrotic core**. Dead tissue cannot exclude calcium (§2.6), so calcium salts nucleate there and grow. The result stains **purple-blue** because haematoxylin is a basic dye and calcium salts are strongly basophilic, which on a section where everything else is pink makes calcification almost impossible to miss. Clinically the calcium matters twice over: it makes the plaque **rigid and brittle**, so the artery cannot dilate and the cap is more likely to crack — and a cracked cap is how a plaque becomes a myocardial infarction.*

PATHOLOGICAL DIAGNOSIS

Coronary atherosclerosis with dystrophic calcification

WRITE IT LIKE THIS

Tissue: Coronary artery. **Description:** The intima is markedly and eccentrically thickened by an atheromatous plaque, reducing the lumen to a narrow crescent (grade IV stenosis). Foam cells are present at the margins of the plaque. The centre of the plaque consists of amorphous necrotic material in which there are scattered irregular granules and lumps of deeply basophilic, purple-blue material. The media beneath the plaque is thinned. **Diagnosis:** Coronary atherosclerosis with dystrophic calcification of the plaque.

PART FIVE

05

Homework, Self-Test and Glossary

What to hand in, the comparisons the lecturer has told you he will examine, eighteen questions to check yourself against, and the vocabulary in one place.

5.1 Your Homework — Exactly What to Submit

Homework-2 is short. Two items, both the same lesion in the same organ, one gross and one microscopic — the pairing is deliberate, because writing both forces you to notice that the yellow greasy organ and the round clear vacuoles are the same fact seen at two scales.

TABLE 5.1 THE SUBMISSION CHECKLIST

#	Item	Task and where it is covered here
1	Gross specimen No. 6 GROSS — Fatty degeneration of the liver	Describe. §3.2 — model answer in the card
2	Slide 10 MICRO — Fatty degeneration of the liver	Describe. §4.3 — model answer in the card

BENCH TECHNIQUE — MAKING THESE TWO REPORTS EARN FULL MARKS

- Use the department's four headings for both items: specimen number and type · organ or tissue · description (evidence of diagnosis) · pathological diagnosis.
- Keep the description in **observable language**. Write “round, sharply outlined, optically clear cytoplasmic vacuoles, some displacing the nucleus”, not “fat”. Save the word fat for the diagnosis line.
- Include one **negative** finding in each. For the slide, “no inflammation, necrosis or fibrosis” is the right one — it shows you considered steatohepatitis and excluded it.
- Because the two items are the same disease, make sure the two descriptions **agree**. A gross description of a yellow greasy liver beside a microscopic description of granular pink cytoplasm is self-contradictory, and a marker will notice.

5.2 Tell Them Apart — the Look-Alikes

TABLE 5.2 PAIRS THAT ARE CONFUSED, AND THE ONE FEATURE THAT SEPARATES THEM

Pair	They share	The deciding feature
Hyaline vs mucoid degeneration	A pale, translucent area in connective tissue	Colour and consistency. Hyaline: white, dense, firm . Mucoid: grey-blue, loose, gelatinous . The lecturer's own correction on specimen 11 makes exactly this point
Hyaline vs calcification (gross)	A grey-white, sharply bounded focus	Touch it. Hyaline is smooth and firm and cuts cleanly; calcium is hard and gritty and catches the knife
Cellular swelling vs fatty change (gross)	Enlarged, soft organ with blunt edges and a tense capsule	Colour. Swelling is pale and dull ; fatty change is yellow and greasy . See §5.3
Reversible vs irreversible injury	Swollen cells with abnormal cytoplasm	Look at the nucleus. Normal nucleus = reversible. Pyknosis, karyorrhexis or karyolysis = the cell is dead

Pair	They share	The deciding feature
Dystrophic vs metastatic calcification	Purple-blue deposits on H&E	Serum calcium, and the tissue. Dystrophic: normal calcium, dead tissue. Metastatic: raised calcium, living tissue
Calcification vs nuclei (micro)	Purple-blue material on a pink section	Shape and place. Nuclei are regular, similar in size, one per cell . Calcium is irregular, amorphous, granular , and sits where no cell should be
Fibroadenoma vs carcinoma (gross)	A breast nodule	The border. Fibroadenoma: oval, well circumscribed, complete capsule , pushes tissue aside. Carcinoma: irregular, infiltrating , no capsule

5.3 The Three Examinable Comparisons

The lecturer states three objectives for Part two of this session. Two of them are comparisons, and the third is a definition set. Here they are, each in one table.

Comparison 1 — cellular swelling versus fatty change

TABLE 5.3 “DISTINGUISH CELLULAR SWELLING AND FATTY CHANGE”

	Cellular swelling	Fatty change
What accumulates	Water	Triglyceride
Immediate cause	Na⁺/K⁺-ATPase failure from ATP depletion	Imbalance between fat delivery, synthesis, oxidation and export
Gross colour	Pale, dull , “like boiling water”	Yellow
Gross feel	Firm and tense	Greasy , soft, light for its size
Vacuole shape	Amorphous , ill-defined	Round , sharply outlined
Vacuole staining	Faintly eosinophilic — contains protein-rich cytosol	Completely clear — the lipid was dissolved out in processing
Nucleus	Central and normal	Pushed to the cell edge by a large vacuole
Special stain	None needed	Sudan III / IV or oil red O on a frozen section — orange-red
This session	Specimen 5, Slides 9 and 46	Specimens 6 and 7, Slide 10

Comparison 2 — tigroid heart versus fat heart

TABLE 5.4 “IDENTIFY TIGROID HEART AND FAT HEART”

	Tigroid heart (No. 7)	Fat heart (No. 8)
Proper name	Fatty degeneration of the myocardium	Fatty infiltration of the myocardium
Where the fat is	INSIDE the myocytes	OUTSIDE the myocytes , between them and in the epicardium
Is it cell injury?	Yes — a metabolic injury to the muscle cell	No — the myocytes are normal
Cause	Hypoxia — severe anaemia, chronic heart failure, toxins (diphtheria)	Obesity ; to a lesser degree ageing
Where you look	Inside — subendocardium of the left ventricle and papillary muscles	Outside — epicardium, especially over the apex
What you see	Orange-red stripes on light brown — a tiger-skin pattern	A thick yellow layer of adipose tissue , extending inwards between muscle bundles
Why that pattern	Cells near capillaries stay oxygenated; cells far from them accumulate fat — and capillaries run in parallel bundles	Fat already normally present in the epicardium simply increases in bulk

Comparison 3 — the three morphologies to master

TABLE 5.5 HYALINE, MUCOID AND CALCIFICATION SIDE BY SIDE

	Hyaline	Mucoid	Calcification
Substance	Protein — different in each site	Mucin (mucopolysaccharide)	Calcium salts (hydroxyapatite)
Gross colour	Grey- white , translucent, frosted glass	Grey- blue , translucent, gelatinous	Grey-white, opaque , fine-grained
Gross feel	Firm, smooth, dense	Soft, slippery	Hard as gravel , gritty
On H&E	Homogeneous deep pink (eosinophilic)	Pale blue-grey , stringy	Deep purple-blue (basophilic), amorphous
Special stain	—	Alcian blue or mucicarmine	Von Kossa or alizerin red
Specimens here	No. 10 pleura · Slide 43	No. 11 fibroadenoma	No. 12 lung · coronary slide

5.4 Self-Test — 18 Questions with Answers

Short-answer, in the format this laboratory is examined in. Cover the answer, produce yours, then compare.

Q1 Give the normal weight of the liver and one feature of its normal edge.

1230–1446 g in a man, 1100–1300 g in a woman. The anterior edge is **sharp**. A blunt, rounded edge is abnormal and appears in both the cloudy-swollen and the fatty liver.

Q2 Name the four structures you identify on a normal liver slide.

Central vein · liver plates · liver sinusoids · portal space (containing the interlobular bile duct, vein and artery).

Q3 How do you distinguish a central vein from a portal vein?

Count the companions. A central vein runs *alone*, with hepatocyte plates coming straight up to its wall. A portal vein sits in connective tissue with *two* companions — a bile duct and an artery. **If you see a duct, you are in a portal tract.**

Q4 Name the four layers of the digestive tube from inside out.

Mucous layer (mucous glands + muscularis mucosae) → **submucosa** (connective tissue) → **muscular layer** (smooth muscle) → **serous membrane** (connective tissue + mesothelium).

Q5 Why does a failing sodium pump make a cell swell?

The Na⁺/K⁺-ATPase normally exports three sodium ions for every two potassium it imports, so stopping it leaves a **net solute load inside the cell**. Sodium accumulates, potassium leaks out, and because the membrane is freely permeable to water, **water follows sodium in** to restore osmotic balance. There is no other possible outcome.

Q6 Why is the proximal tubule the commonest site of hydropic change in the kidney?

It is the **most metabolically active** segment — it reabsorbs most of the filtered water, sodium, glucose and amino acids. That work demands a great deal of ATP and a great many mitochondria, which makes the cell **unusually sensitive to hypoxia, poisoning and infection**. When energy metabolism fails, its sodium pump fails first.

Q7 Why is a cloudy-swollen liver pale and dull rather than red and glistening?

Pale because the swollen hepatocytes compress the sinusoids and squeeze blood out of the tissue. **Dull** because the dilated endoplasmic reticulum and swollen mitochondria scatter light instead of transmitting it — the same reason a poached egg white is opaque, which is what the phrase “resembling boiling water” is describing.

Q8 Give four mechanisms by which fat accumulates in a hepatocyte, with a cause for each.

Increased delivery of free fatty acids (starvation, diabetes, obesity) · **increased synthesis** (alcohol) · **decreased β -oxidation** (hypoxia, alcohol, toxins) · **decreased apoprotein synthesis** so triglyceride cannot be exported as VLDL (carbon tetrachloride, protein malnutrition).

Q9 Which zone of the liver lobule shows fatty change first in hypoxia, and why?

Zone 3 — centrilobular, around the central vein. Blood enters at the portal triads and leaves at the central vein, so zone 3 receives blood already stripped of oxygen by the cells upstream and is therefore the **least oxygenated** and the first to fail.

Q10 On an H&E slide, how do you tell a hydropic vacuole from a fat vacuole?

Four features. **Shape:** hydropic is amorphous, fat is round. **Edge:** hydropic is ill-defined, fat is sharply outlined. **Colour:** hydropic is faintly eosinophilic (it contains protein-rich cytosol), fat is completely clear (the lipid was dissolved out). **Nucleus:** hydropic leaves it central, fat pushes it to the cell edge.

Q11 Why are the fat vacuoles empty on your slide, and how would you show the fat?

Routine processing uses **organic solvents that dissolve lipid away**, so you are looking at the holes it left. To show it, avoid the solvents: cut a **frozen section** and stain with **Sudan III, Sudan IV or oil red O**, which turn lipid orange-red.

Q12 What is the single feature that tells you a swollen cell is still alive?

A normal nucleus. Even a fully ballooned cell is reversibly injured as long as its nucleus is intact. Pyknosis, karyorrhexis or karyolysis means the cell is dead and the process is necrosis, not degeneration.

Q13 Why is a tigroid heart striped?

The stripes are a **map of the oxygen supply**. Myocytes close to a capillary stay oxygenated and metabolise their fatty acids; those furthest away cannot complete β -oxidation and accumulate triglyceride. Since the capillaries run in **parallel bundles**, the pale fat-laden bands alternate with normal red-brown ones — a tiger stripe.

Q14 A heart is covered in yellow fat. Is that a tigroid heart?

No — that is a fat heart. Tigroid heart is fatty **degeneration**: fat *inside* the myocytes, seen as orange-red stripes *beneath the endocardium*, caused by hypoxia. Fat heart is fatty **infiltration**: excess adipose tissue *outside* the myocytes, in the epicardium and between muscle bundles, caused by obesity. The myocytes in a fat heart are entirely normal.

Q15 What is hyaline made of?

It depends where it is. “Hyaline” is a descriptive term for anything homogeneous, glassy and eosinophilic, not a single substance. Fused collagen in a scar or thickened pleura; plasma protein and basement-membrane material in an arteriolar wall; immunoglobulin in a plasma cell (Russell body); keratin in a hepatocyte (Mallory body). Four unrelated substances, one appearance.

Q16 In specimen 11, which is the hyaline and which the mucoid area?

The lecturer corrects this explicitly: the **white stripe is hyaline degeneration**, and the **irregular grey-blue area is mucoid degeneration**. In general: white, dense and firm is hyaline; grey-blue, loose and gelatinous is mucoid.

Q17 Distinguish dystrophic from metastatic calcification.

Dystrophic: calcium deposited in **dead or dying tissue**, with a **normal serum calcium** — the problem is local. Caseous necrosis, atheromatous plaque, old infarct, damaged valve. **Metastatic:** calcium deposited in **normal living tissue** because the **serum calcium is raised** — the problem is systemic. Hyperparathyroidism, bone destruction, vitamin D excess, renal failure; it favours acid-secreting tissues such as gastric mucosa, kidney and lung. Both specimens in this session are dystrophic.

Q18 Why does calcification look purple-blue on an H&E slide?

Because **haematoxylin is a basic dye** and **calcium salts are strongly basophilic**, so they bind it avidly. On a section where almost everything else is pink, an **irregular, amorphous, granular purple-blue deposit that is not a nucleus** is calcium until proved otherwise.

5.5 Glossary of Key Terms

Alcian blue

Special stain for acid mucins, which it turns blue. Used to confirm that a grey-blue gelatinous area really is mucoid degeneration.

Amorphous

Without definite shape. Applied to the hydropic vacuole, it is the word that separates it from the round, sharply outlined fat vacuole.

Anthraxis

Black pigmentation of lung from inhaled carbon. Normal in any adult city-dweller — and the black surround of the calcified lesion in specimen 12.

Balloon cell

A hepatocyte at the most severe reversible stage of hydropic change: hugely distended and almost transparent, but with a **normal nucleus**. That last point is what keeps it on the reversible side of the line.

Basophilic

Taking up the basic dye haematoxylin and appearing blue-purple. Nuclei and calcium salts are basophilic; almost everything else on an H&E slide is not.

Caseous necrosis

Cheese-like necrosis characteristic of tuberculosis. It is dead tissue, so it is a classic site for **dystrophic calcification** — which is the whole story of specimen 12.

Cellular swelling

Accumulation of water inside a cell after failure of the ATP-dependent sodium pump. Called **cloudy swelling** grossly and **hydropic degeneration** microscopically. The first manifestation of almost all cell injury, and fully reversible.

Centrilobular (zone 3)

The region of the hepatic lobule around the **central vein** — the last to receive blood and therefore the **least oxygen-**

ated. First and worst affected in any hypoxic injury to the liver.

Cloudy swelling

The gross appearance of cellular swelling: the organ is **enlarged, pale, dull and tense**, with rounded edges and a bulging cut surface. The lecture's phrase is "resembling boiling water".

Dystrophic calcification

Calcium deposited in **dead or dying tissue** while the **serum calcium is normal**. The problem is local. Both calcification specimens in this session are of this kind.

Eosinophilic

Taking up the acidic dye eosin and appearing pink. Cytoplasm, collagen and hyaline material are eosinophilic.

Fat heart

Fatty **infiltration** of the heart: an excess of adipose tissue in the epicardium and between muscle bundles. The fat is **outside** the myocytes and the myocytes are normal. Caused by obesity. Contrast *tigroid heart*.

Fatty change (steatosis)

Abnormal accumulation of **triglyceride inside parenchymal cells**, most often hepatocytes. Grossly the organ is enlarged, yellow, greasy, soft and light; microscopically the cells contain round, clear vacuoles of varying size, a large one displacing the nucleus.

Foam cell

A macrophage stuffed with lipid, so called because the dissolved-out lipid leaves its cytoplasm frothy. Found at the margins of an atheromatous plaque; its death contributes to the necrotic core.

Hyaline degeneration

Any accumulation that appears **homogeneous, glassy, translucent and eosinophilic**. From Greek *hyalos*, glass. A **description, not a substance** — the material is different in connective tissue, in a vessel wall and inside a cell.

Hydropic degeneration

The microscopic form of cellular swelling: loose, granular, reticulate or transparent cytoplasm with ill-defined, faintly eosinophilic vacuoles, and a **normal nucleus**.

Hydroxyapatite

The crystalline calcium-phosphate mineral of bone — and of pathological calcification. Once formed it is permanent; the body has no mechanism for removing it.

Macrovesicular steatosis

Fatty change in which a **single large vacuole** fills the hepatocyte and flattens the nucleus against the membrane, so the cell resembles an adipocyte. The pattern of alcohol and of metabolic syndrome.

Mucoid (myxoid) degeneration

Accumulation of **mucin in connective-tissue stroma**, making it grey-blue, translucent, loose and gelatinous. Seen in mesenchymal tumours such as fibroadenoma, and in myxoedema.

Na⁺/K⁺-ATPase

The membrane pump that exports three sodium ions for every two potassium it imports, at the cost of one ATP. Its failure is the first step of cellular swelling and consumes a large share of a cell's energy budget.

Portal space (portal triad)

The connective-tissue island at the corner of a hepatic lobule containing **three** structures: the interlobular bile duct, vein and artery. Marks zone 1 — periportal, best oxygenated.

Pyknosis · karyorrhexis · karyolysis

The three nuclear changes of cell death: shrinkage and condensation, fragmentation, dissolution. Their **absence** is what makes an injury reversible.

Reversible injury

Cell injury from which the cell can fully recover if the cause is removed. Morphologically: **cellular swelling and fatty change**, with a normal nucleus throughout. The subject of this session.

Sinusoid

The wide, irregular capillary between hepatic plates. **Narrowing of the sinusoids** is the most reliable low-power sign that hepatocytes have enlarged, in both swelling and fatty change.

Sudan III

A fat-soluble dye that partitions into lipid and stains it orange-red. Must be used on a **frozen section**, because routine processing dissolves the fat away. Specimen 6 was tested this way and was positive.

Tigroid heart

Fatty **degeneration** of the myocardium: orange-red stripes on a light brown background beneath the left ventricular endocardium and papillary muscles, resembling tiger skin. The fat is **inside** the myocytes, and the cause is hypoxia.

Von Kossa · alizarin red

Special stains for calcium: von Kossa blackens it, alizarin red turns it orange-red. Rarely needed, because calcium is already unmistakable on H&E.